A Multiply-Less Approximate SRAM Compute-In-Memory Macro for Neural-Network Inference

Haikang Diao, *Graduate Student Member, IEEE*, Yifan He, *Graduate Student Member, IEEE*, Xuan Li, Chen Tang, *Graduate Student Member, IEEE*, Wenbin Jia, Jinshan Yue, *Member, IEEE*, Haoyang Luo, *Graduate Student Member, IEEE*, Jiahao Song, *Member, IEEE*, Xueqing Li, *Senior Member, IEEE*, Huazhong Yang, Hongyang Jia, *Member, IEEE*, Yongpan Liu, *Senior Member, IEEE*, Yuan Wang, *Member, IEEE*, and Xiyuan Tang, *Senior Member, IEEE*

***Abstract*— Compute-in-memory (CIM) is promising in reducing data movement energy and providing large bandwidth for matrix-vector multiplies (MVMs). However, existing work still faces various challenges, such as the digital logic overhead caused by the multiply-add operations (OPs) and structural sparsity. This article presents a 2-to-8-b scalable approximate digital SRAM-based CIM macro co-designed with a multiply-less neural network (NN) approach. It incorporates dynamic-logic-based approximate circuits for the logic area and energy saving by eliminating multiplications. A prototype is fabricated in 28-nm CMOS technology and achieves peak multiply-accumulate (MAC)-level energy efficiency of 102 TOPS/W for 8-b operations. The NN model deployment flow is used to demonstrate CIFAR-10 and ImageNet classification with ResNet-20 and ResNet-50 style multiply-less models, respectively, achieving the accuracy of 91.74% and 74.8% with 8-bit weights and activations.**

***Index Terms*— Compute-in-memory (CIM), dynamic logic, multiply-less approximate, neural network (NN) inference.**

I. Introduction

ARTIFICIAL intelligence based on neural network (NN) has enabled various emerging applications, ranging from the edge to cloud computing, such as computer vision, language processing, and molecular discovery for scientific applications [1], [2], [3], [4]. A key attribute of those applications is that they heavily rely on computation with huge data, such as high-dimensional matrices and tensors. Thus, the efficiency and performance of the NN inference, which is dominated by matrix-vector multiplies (MVMs), are limited by the memory access and I/O bandwidth [5]. However, due to the high-dimensionality computation in MVMs, Von Neumann architecture-based computing platforms are not well adapted to NN inference. A new hardware solution needs to be proposed to advance modern AI development.

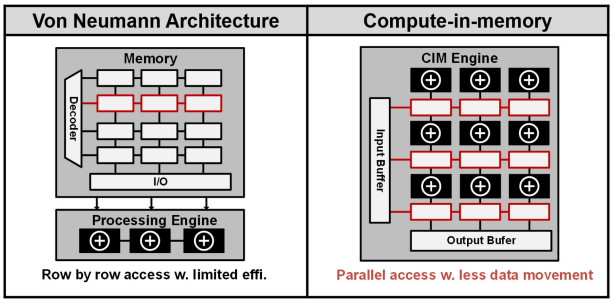


Fig. 1. Difference between Von Neumann architecture and CIM.

As shown in Fig. 1, computing-in-memory addresses this limitation by incorporating the processing unit into memory cells to perform basic operations (OPs) locally [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. Thus, the MVMs can be finished efficiently due to reduced weight movement. Moreover, multiple rows can be activated simultaneously in parallel based on the input vector to provide large bandwidth. Analog compute-in-memory (CIM) implement multiply-accumulate (MAC) in current [6], [7], [8], [9], [27] and charge domains [10], [11], [12], [13]. They are distinguished by circuit structures that perform multiplication and dimension reduction operations. The current-domain CIM relies on the summation of modulated currents through the bit-cell transistors. Although achieving high density, it suffers from nonidealities like transistors PVT variations. The charge-domain CIM realizes multiplication by the precharging of capacitors and accumulation by charge redistribution. However, both solutions require ADCs for circuit readout,

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Haikang Diao, Haoyang Luo, and Xiyuan Tang are with the Institute for Artificial Intelligence and the School of Integrated Circuits, Peking University, Beijing 100871, China (e-mail: diaohaikang@stu.pku.edu.cn; xitang@pku.edu.cn).

Yifan He, Chen Tang, Wenbin Jia, Xueqing Li, Huazhong Yang, Hongyang Jia, and Yongpan Liu are with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China.

Xuan Li is with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China, and also with Wiener Technology, Beijing 100080, China.

Jinshan Yue is with the Institute of Microelectronics of the Chinese Academy of Sciences, Beijing 100029, China.

Jiahao Song is with the Institute for Neural Computation (INC), University of California at San Diego, La Jolla, CA 92093 USA.

Yuan Wang is with the School of Integrated Circuits, Peking University, Beijing 100871, China.

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significantly increasing the power and area consumption, especially with higher bit-widths.

Digital CIM is proposed recently [14], [15], [16], [17], [18], [19], [20], [21], [22]. It performs multiplication and dimension reduction by digital logic, offering exceeding robustness and easier architectural integration. The first work to present digital CIM, as detailed in [16], adopts hybrid adders to support full-precision MAC operations and enhance efficiency. After that, the 12-T SRAM design is implemented in 5-nm technology with the support of simultaneous write and CIM operations [17]. Then, the dynamic logic is introduced for various types of bit-wise operations with reduced circuit area [20]. However, the multipliers and adder trees adopted in digital CIM consume a significant area and energy, limiting their performance.

To reduce digital circuit costs, recent works have started to incorporate approximate computation. For example, an approximate digital CIM macro that uses OR/AND digital logic circuits to replace multiple levels of adders in the adder tree is proposed in [15]. It reduces the transistor count of digital circuits by 56%. Nevertheless, this approximation results in a 3.5% accuracy loss on the Cifar-10 dataset, limiting its usage in advanced applications. As a result, the current approximate digital CIM design improves area and energy efficiency at the cost of accuracy.

In addition to using approximations to reduce the computation cost in digital CIM, the block- and data-level sparsity is also exploited to reduce the number of computations for further energy efficiency boost [21], [22]. However, it requires extra sparse training, together with additional hardware, to process this coarse-gain sparse information.

This work proposes an approximate digital CIM macro co-designed with a multiply-less NN approach, incorporating an architecture design dedicated to multiply-less NN, an L1-distance computing scheme, and dynamic-logic-based approximate circuits for multiply-less NN inference. Compared to prior designs, this work employs co-optimization from the algorithm to the circuit level, reducing the approximation loss for NN inference. The primary contributions of this work are listed as follows.

1) This work proposed an efficient architecture for a multiply-less NN that approximates dot products with L1-distance of vectors to save area and energy consumption caused by multiplication.
2) This work proposed an L1-distance computing scheme that reformates bit-parallel absolute-difference operators in the L1 compute into bit-serial comparisons and guarantees linear area/energy scaling with bit precision.
3) This work proposed a dynamic-logic-based comparator design with early weight-bit-readout stopping that exploits unstructured bit statistics. Compared to the conventional static logic-based design, the proposed dynamic design further reduces area and energy consumption.

In addition, an NN model deployment flow is proposed to implement the custom off-line posttraining quantization adapted to chip architecture. Multiple representative NN benchmarks are demonstrated on the chip, achieving accuracies equivalent to ideal digital computation.

This article is an extension of [19] with an improved dynamic-logic-based comparator design. The improved comparator greatly reduces the comparison error rate and improves the accuracy of NN inference. This design was fabricated, measured, and compared with [19]. The rest of this article is organized as follows. Section II discusses the challenges of approximate digital CIM. Section III outlines the efficient architecture for multiply-less NN algorithms. Section IV describes the efficient computing scheme for adder-based approximation. Section V presents the efficient comparator circuits with dynamic logic. Section VI presents the prototype measurements and NN demonstrations. The conclusion is drawn in Section VII.

II. Challenges of Approximate Digital CIM

The first challenge is the computational error introduced by circuit-level approximation. Fig. 2(a) shows the two circuit-level approximation methods for the adder tree: the OR-only approximation and the interleaved OR/AND approximation [15]. Both approaches use simple logic gates to replace multiple levels of adders in the adder tree. S1, S2, and S3 denote the approximation of adders with single/double/triple stages in adder tree. For example, S2 indicates that a four-input logic gate is used instead of a two-stage adder to achieve four-input summation. However, the errors introduced by the approximate computation degrade the NN accuracy as efficiency increases. Unlike the previous circuit-level approximation, this work incorporates algorithmic-level approximation to reduce errors for better NN inference accuracy.

The second challenge is the quadratic cost of multiplications introduced by the expansion of bit-widths in digital CIM with input bit-level parallelism. Although bit-serial digital CIMs reduce this cost by simplifying multipliers to NOR gates [16], it requires greatly increased compute cycles. As shown in Fig. 2(b), the area-cycle product, an area efficiency indicator, increases quadratically with bit-widths, regardless of bit-parallel or bit-serial operations. This quadratic-growth multiplication cost limits further area efficiency improvements for digital CIM. To overcome this limitation, this work implements NN inference using multiply-less matrix-vector operations, ensuring a linear area/energy scaling with bit-widths.

The third challenge falls into the insufficient usage of bit-level sparsity of NN models. Several works have proposed sparsity-aware designs for energy efficiency improvement. Their works focus on the block or data level sparsity. In [21] and [24], consecutive zeros in the weights or activations can be skipped during computation by gating the corresponding readout circuits. Compared to the coarse-grained sparsity, as shown in Fig. 2(c), the NN model naturally shows higher finer-grain sparsity at the bit-level. Specifically, in a typical convolutional NN (CNN) layer without sparse-aware training, the bit-level statistics show an unbalanced binary distribution leaning heavily to zero, which has not been fully exploited in previous digital CIMs due to the tight and structural memory layout. To leverage the sparsity of the bit-level distribution, this work designs specialized circuits and NN deployment

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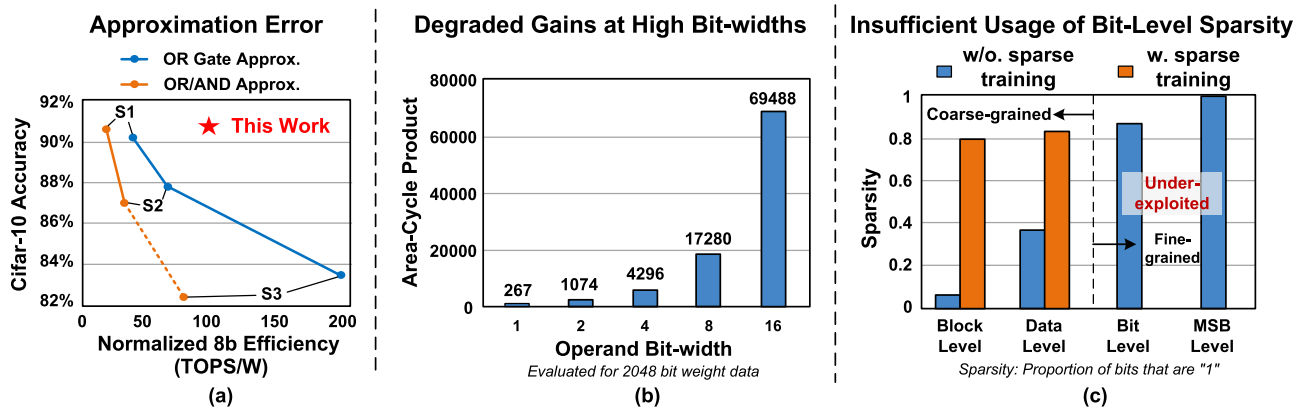


Fig. 2. Three challenges of approximate digital CIM. (a) Approximation error by circuit-level approximation. (b) Degraded gains due to multiplications at high bit-widths. (c) Insufficient use of bit-level sparsity.

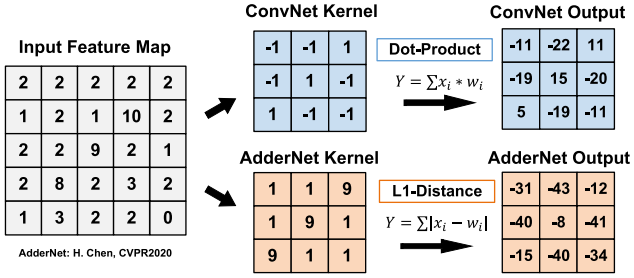


Fig. 3. Example for AdderNet kernel: achieves similar output without multiplication.

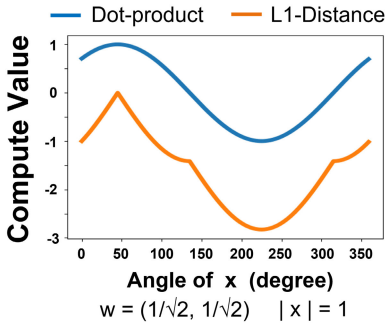


Fig. 4. Similarity between ConvNet (MAC) and AdderNet (L1-distance).

dataflow for further energy reduction. This design does not require specialized sparse training to adapt the dataflow, and the versatility is improved.

III. Efficient Architecture for Multiply-Less Neural Network Algorithm

A. Background of Multiply-Less NN Algorithm

To mitigate the large area and cycle penalty of multiplication in deep learning inference, recent algorithms start to explore all addition NN layers. For example, AdderNet replaces traditional convolutional kernels with adder-based kernels [28], [29], [30], [31], [32], [33]. It uses the L1-distance to replace the dot-product between weights and activations

$$Y(m,n,t) = -\sum_{i=0}^{d}\sum_{j=0}^{d}\sum_{k=0}^{c_{\text{in}}} |X(m+i,n+j,k) - W(I,j,k,t)|. \quad (1)$$

The expressive power of L1-distance operations was quantitatively analyzed. A formula derivation in [34] shows that L1-distance has the same approximation capacity as traditional CNN. As shown in Fig. 3, both ConvNet kernel and AdderNet kernel can detect and strengthen the diagonal edge in the input feature map. Fig. 4 shows the similarity between AdderNet and CNN kernels. When the weight unit vector is fixed and the input unit vector is rotated, the results of the dot-product and L1-distance demonstrate similarity. Therefore, for NN inference, the CNN kernel can be approximated by the AdderNet kernel and the hardware design of AdderNet can benefit from this simple computation. Table I shows the inference performance of ConvNets and AdderNets, and the effectiveness of AdderNet has been validated on different models such as CNN and transformer, and different tasks such as image classification and object detection [28], [29], [30], [31], [32], showing comparable performance and scalability to traditional CNN.

TABLE I
Performance Comparison of ConvNets and AdderNets

Task	Image Classification on ImageNet dataset			Object Detection on COCO dataset
Model	ResNet-18 (CNN)	ResNet-50 (CNN)	DeiT-B (Transformer)	R-CNN
Multiply-based	69.8%	76.2%	81.8%	37.9% (mAP)
AdderNet	67.0%	74.9%	80.4%	37.0% (mAP)

B. Architecture for Approximate Compute

The macro architecture for approximate computation is shown in Fig. 5, with three features highlighted in different colors. It comprises 16 Digital Approximate CIM Arrays for L1-distance-based approximate MVMs, an activation buffer (ACT buffer) for receiving and sequencing activations to the CIM array, an activation accumulator (ActACCU) for 128 activation accumulation per cycle, a standard SRAM read/write interface controller, and other peripherals for CIM control.

Each digital approximate CIM array contains 8 (rows) × 128 (columns) 8T-SRAM array for NN weights storage, natively supporting input vector dimension of 128, and

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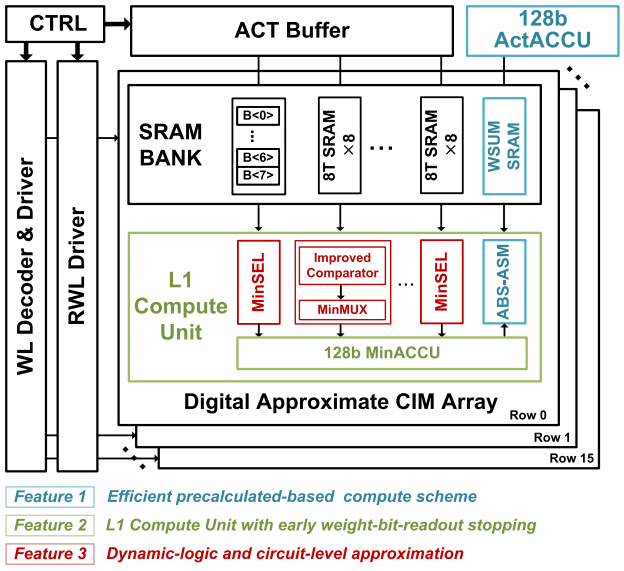
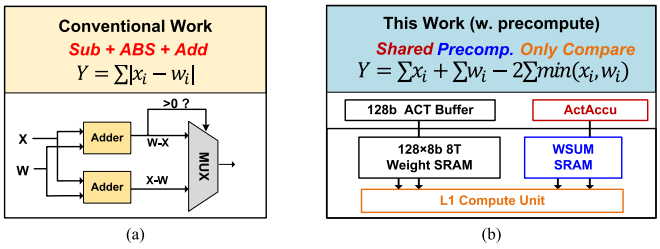


Fig. 5. Overall architecture of multiply-less approximate CIM macro.



(a) (b)

Fig. 6. (a) Conventional L1-distance computing scheme. (b) Recalculation-based L1-distance computing scheme.

2-to-8-b precision. The SRAM array is associated with an L1 computate unit for comparison-based computation. The L1 computate unit contains minimum-value selection (MinSEL), absolute-value-assembly unit (ABS-ASM), and minimum-value accumulator (MinACCU), using bit-serial dataflow to enable linear bit-scaling cost and leverage bit-distribution. The MinSEL contains an improved dynamic-logic-based approximate bit-serial comparator and minimum-value-multiplexer (MinMUX), which are used to implement the dynamic-logic comparison.

IV. Compute Scheme for Adder-Based Approximation

A. Precalcultion-Based L1-Distance Compute Scheme

The L1-distance computing consists of three steps: subtraction of weights and activations, taking absolute values, and accumulation. The conventional L1-distance computing scheme is shown in Fig. 6(a), which uses two multibit adders and a multiplex to calculate the absolute difference between weights and activations [35]. Although the overhead of these digital logics is reduced compared to dot-product, the multibit adders still limit the area efficiency. Moreover, due to the absolute values required before accumulation, a bit-serial computing scheme would not be available, making it difficult to exploit the sparsity of the bit distribution.

In this article, we proposed an efficient precalculation-based L1-distance compute scheme that is employed in digital approximate CIM Array. It simplifies the conventional absolute difference computation to a simple minimum selection. As shown in Fig. 6(b), the absolute value of the L1-distance can be expanded and rewritten as a summation of activations and weights, and a minimum selection between activations and weights. In the new expression, the summation of weights (WSUM), $\sum w_i$, is stationary and can be precomputed during model compilation and stored locally using additional columns in each SRAM bank, consuming only 15 additional SRAM cells. While the summation of activations, $\sum x_i$, is handled by a dedicated ActACCU succeeding the ACT Buffer, whose result is shared across all 16 digital approximate CIM arrays to amortize the extra computation costs. Last, only the minimum summation, $2\sum \min(x_i, w_i)$, needs to be computed by the L1 compute unit. Comparing with the conventional absolute difference computation, the proposed design achieves 1.5× area reduction by replacing the subtraction and absolute value with comparison.

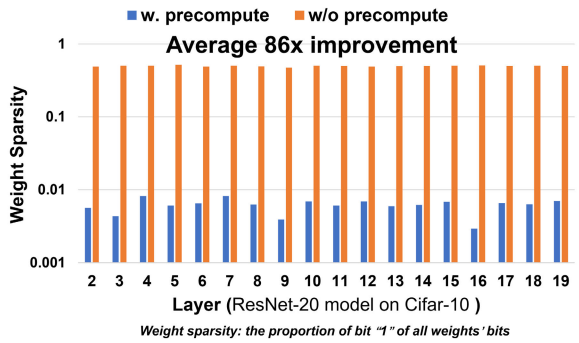


Fig. 7. Weight sparsity gains due to the precomputation.

Another benefit of the precomputation is that it significantly improves the effective sparsity of the weight tensor stored in the macro. The nonnegative character of the rectified linear unit (ReLU)-based activations enables the truncation of negative weights to zero in the minimum selection function

$$\sum|x_i - w_i| = \sum|x_i - \text{clamp}(w_i, 0, w_{\max})| + \sum|w_i|. \quad (2)$$

The equivalent calculation is achieved by merging the truncated weights into the WSUM. Fig. 7 shows the weight sparsity gain due to the precomputation. For the ResNet-20 style multiply-less model on the Cifar-10 dataset, the sparsity of the weights is increased by 86× with precomputation compared to that of the original model. Higher sparsity can effectively reduce the activity factor in the L1 compute unit, reducing the power consumption of comparison operations. Moreover, as the weights and activations are nonnegative, sign bits or complements are not required. Therefore, the bit-widths are reduced, and the circuit design is simplified.

B. Bit-Serial Comparison Scheme

The circuit-level design and bit-serial computation of L1 compute unit are shown in Fig. 8. The L1 compute unit sums the minimum values and then combines the sum of minimum values, the sum of weights, and the sum of activations ($\sum x_i$) using an ABS-ASM. The bit-serial dataflow is adopted

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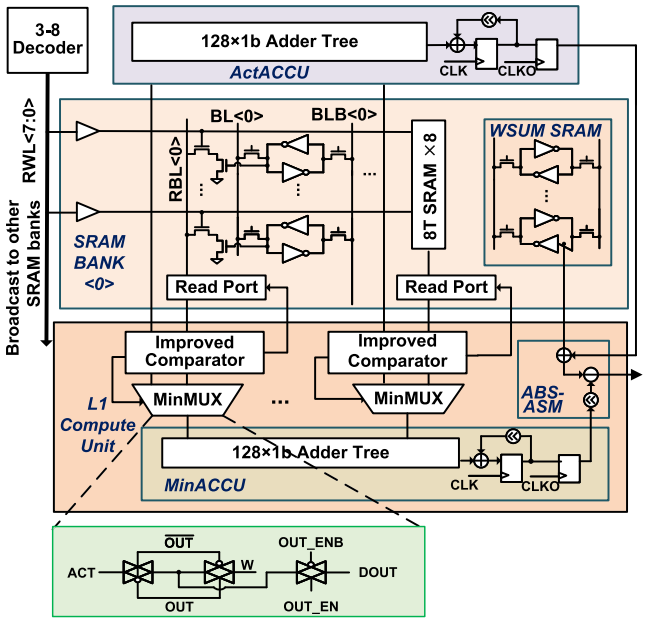


Fig. 8. Circuit-level design and bit-serial computation of L1 compute unit.

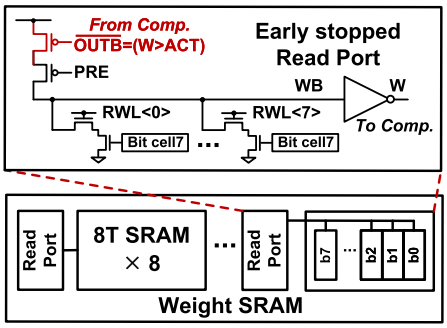
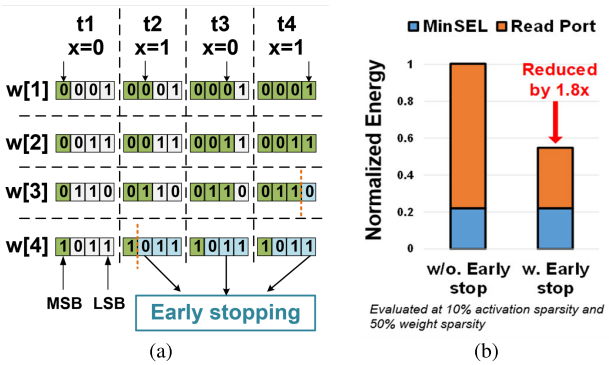


Fig. 9. Schematic of the specialized local SRAM read port.



(a) (b)

Fig. 10. (a) Early weight-readout stopping mechanism. (b) Read port energy saving for the early stop.

to support different bit-width in the L1 compute unit. The activations are serially sent into the macro in the MSB-first manner. After the ACT Buffer, the activations will be shared between ActACCU and 16 digital approximate CIM arrays. The ActACCU shifts and accumulates one bit of activations per cycle, and broadcasts the result to different L1 compute units. The weight SRAM bank contains a 128 × 8 SRAM bit cells subarray and 15 extra SRAM bits for weight sum. The RWL decoder selects the weight bit in the MSB-first manner, in the same order as the activations input. The corresponding weights and activations are sent to the Improved Comparator for the comparison result.

The output of the comparison results controls the precharging of the SRAM read port for early stopping. Thus, it reduces the power consumption of the SRAM reading. More details will be provided in Section IV-C. Moreover, the comparison results also control the transmission gate in MinMUX to select the minimum of the weights and activations. Fig. 8 shows the circuit-level details of the MinMUX. The OUT and $\overline{\text{OUT}}$ nodes represent the result of the bit-serial comparison. Initially, OUT is precharged to VDD, and $\overline{\text{OUT}}$ is discharged to VSS. MinMUX passes the weight bit until the comparison result is determined. If the minimum value is the weight, then OUT and $\overline{\text{OUT}}$ are fixed, and MinMUX will keep passing the weight bit until all bits are compared. In contrast, if the minimum value is activation, OUT is discharged to VSS and $\overline{\text{OUT}}$ is charged to VDD, and the MinMUX will keep passing the activation bit.

The outputs of MinMUX are sent to the MinACCU, which shares the same structure as ActACCU, for the summation of the minimum values. Both the ActACCU and MinACCU have clock-gated output to reduce the output activity. The $\sum w_i$ are read out in parallel from the internal nodes of the WSUM SRAM bit cells. Finally, the ABS-ASM collects the output from the ActACCU ($\sum x_i$), MinACCU [$2\sum \min(x_i, w_i)$], and the WSUM SRAM ($\sum w_i$) to generate the partial sums.

C. Sparse-Aware Comparison With Early Stop

The bit-serial minimal selection scheme gives us opportunities to benefit from bit-level distribution. Fig. 9 shows the circuit-level details for the read port of 8T SRAM, which contains a specialized local SRAM read port. When the selected weight bit is "0," the local read bitline will be maintained at the precharged voltage. Therefore, when the bit distribution of weights contains consecutive "0," it will reduce the power consumption of SRAM reading. As mentioned above, by truncating the weights of negative values to zero in the precomputation process, the number of "0" in the bit distribution of the weights is substantially increased, and the power consumption of the SRAM reading is further reduced.

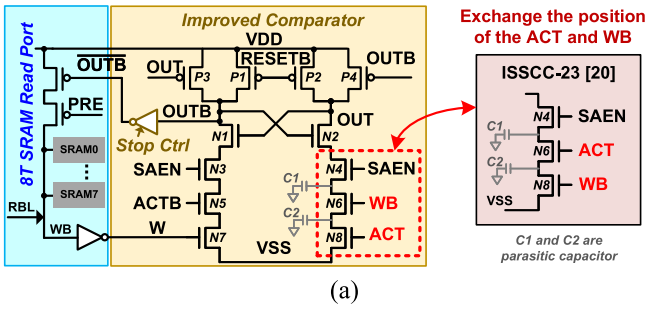
In addition, once the activation is selected, $\overline{\text{OUT}}$ is charged to VDD. The read port stops the repetitive precharging of the local read bitline, allowing an early stopping to reduce power consumption. As shown in Fig. 10(a), the weight bits, which are colored blue, do not need to be precharged and read again. In the later cycles, the MinMUX directly outputs the activation bits to the MinACCU. As shown in Fig. 10(b), with the special design of the read port, the power consumption of the circuit can be reduced by 1.8×.

V. Energy Efficient Comparator Design

A. Improved Dynamic-Logic-Based Bit-Serial Comparator

As shown in Fig. 11(a), the improved dynamic-logic-based approximate bit-serial comparator (Improved Comparator) for bit-serial comparison is proposed to improve the energy efficiency. $C1$ and $C2$ are parasitic capacitors under nodes between $N4$, $N6$, and $N8$. The output nodes OUT and OUTB are precharged to VDD by RESETB at the initial state. If the

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Bit-Serial MSB to LSB		Last Cycle	This Cycle
ACT	W	OUT/ OUTB	OUT/ OUTB
Don't Care		0/1	0/1
		1/0	1/0
0	0	1/1	1/1
0	1		1/0
1	0		0/1
1	1		1/1

(b)

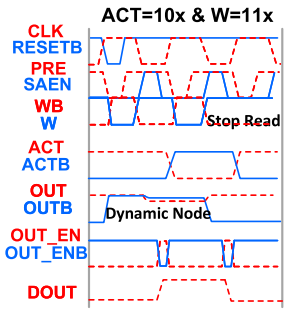


Fig. 11. (a) Schematic of the improved comparator. (b) State transition table for improved comparator. (c) Waveform for comparison operation.

activation and weight bits are equal in the current cycle, the OUT and OUTB hold the charge and pass the equal state to the next cycle. Once the comparison is asserted, one node is discharged to VSS, and the other is pinned to the VDD level through $P3$ or $P4$. Specifically, OUT will be discharged to VSS when the weight is smaller, and conversely, OUTB will be discharged when the activation is smaller. Compared to the original comparator design in [19], the improved comparator in this article only exchanges the positions of ACT and WB, but significantly reduces the comparison error rate. This will be analyzed in detail in Section V-B. Fig. 11(b) summarizes the state transition truth table for the improved comparator.

Fig. 11(c) presents an example of the comparison operation. First, RESETB is set to VSS to precharge OUT and OUTB nodes. After that, RESETB is set to VDD until all bits are compared. Second, activations and weights are serially sent to the improved comparator for dynamic comparisons. Note that each time the RBL is precharged to read the weights stored in the SRAM, the RBL voltage is changed, affecting the W and WB in the improved comparator. Therefore, during this phase, SAEN needs to be discharged to VSS to prevent the voltage change of the RBL from incorrect comparison results. Finally, when the comparison asserted (ACT $<$ W) in the second bit of the weight and activation, OUT connects to VDD, and OUTB connects to VSS. After that, the voltages of OUTB and OUT will remain constant until all the bits have been compared. As a comparison, a bit-serial static-logic-based comparator is designed by the digital flow design (Verilog coding in VS Code, digital synthesis by Design Compiler, and digital backend by Innovus), and its area and power consumption are postlayout simulated using virtuoso. As shown in Fig. 12, compared to the conventional static-logic-based comparator, the improved comparator reduces the area and power by 1.7× and 4.1×, respectively.

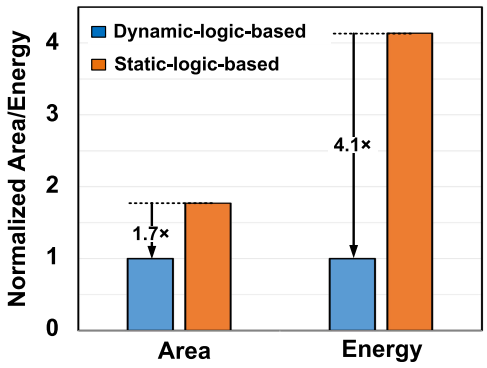


Fig. 12. Area and energy comparison between the static-logic-based comparator and the dynamic-logic-based comparator.

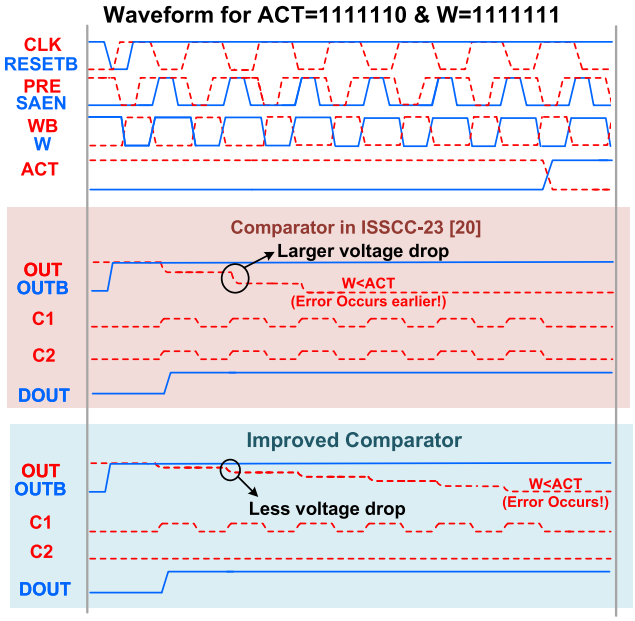


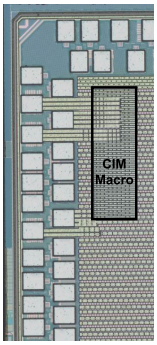
Fig. 13. Comparison errors for the comparator in [19] and improved comparator.

B. Error Analysis for Dynamic Comparator

As discussed, the proposed improved comparator can reduce the potential accuracy loss caused by the original comparator design [19]. To fully understand this mechanism, the error source of the comparator in [19] is first analyzed. The original design suffers from the charge-sharing problem, especially when the activation and weight bits are both "1" for multiple cycles. Fig. 13 shows an example of comparison errors. First, the OUT and OUTB nodes are precharged to VDD by setting RESETB to VSS. For the first bit, ACT and W are both "1," and ACTB and WB are both "0." Thus, $N7$ and $N8$ are turned on, and $N5$ and $N6$ are turned off. When the first-bit comparison is performed, SAEN is charged to VDD. Since both $N4$ and $N6$ are turned on, parasitic capacitors $C1$ and $C2$ are performed charge-sharing with OUT, resulting in a large voltage drop at OUT. After that, when reading the second bit of the weights, the RBL is precharged to VDD and SAEN is discharged to VSS, causing $N8$ to be turned on and $N4$ to be turned off. This results in the $C1$ and $C2$ nodes being discharged to VSS. Notice that when it performs the second-bit comparison, $C1$ and $C2$ are performed charge sharing with OUT again, resulting in a continuous voltage drop at

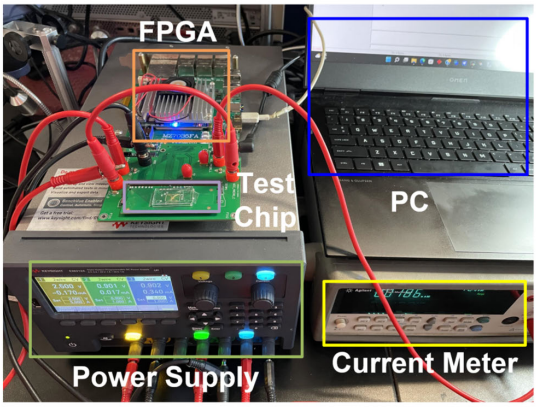
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Chip Summary	
Technology	28nm
Macro Area[1]	0.028mm^2
SRAM Capacity	16Kb
Supply Voltage	0.54-0.9V
NN Operand	ABS+ADD
Clock Frequency	20-240MHz
ACT/W Precision	2-8b
8b Throughput[2]	0.122 TOPS
8b Energy Efficiency[3]	102 TOPS/W



[1]: The In/Out SRAM and pre-computation module for weight summation are excluded
[2]: Measured at 0.9V, 230MHz, 50% input bit sparsity, 50% weight sparsity
[3]: Measured at 0.54V, 20MHz, 10% input bit sparsity, 50% weight sparsity

(a)



(b)

Fig. 14. (a) Chip micrograph. (b) Test platform.

OUT. According to the simulation, after three consecutive comparisons, a comparison error occurs.

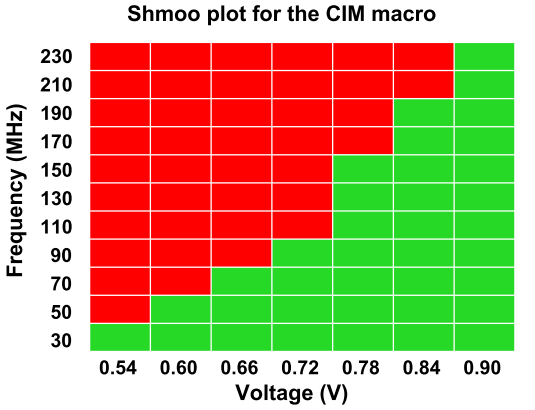
This article proposed an improved comparator to reduce the voltage drop at OUT by exchanging the positions of ACT and WB. Specifically, when the first-bit comparison is performed, $N4$ is turned on, and $N6$ is turned off, and only $C1$ is performed charge-sharing with OUT, resulting in less voltage drop at OUT. As shown in Fig. 13, the error does not occur until after six consecutive comparisons. Thus, compared with [19], the comparator proposed in this work can reduce the voltage drop due to charge sharing, which in turn largely reduces the comparison error rate.

To summarize, the comparison may be contaminated when the number of high ACT-W pairs exceeds a threshold. This approximation error can be avoided by regularizing the maximum number of "1"s in activations during NN inference, with lower bits truncated. The proposed improved comparator relaxes the activation regulation compared to the original design in [19], thus improving the application accuracy, which will be explained in detail in Section VI-A.

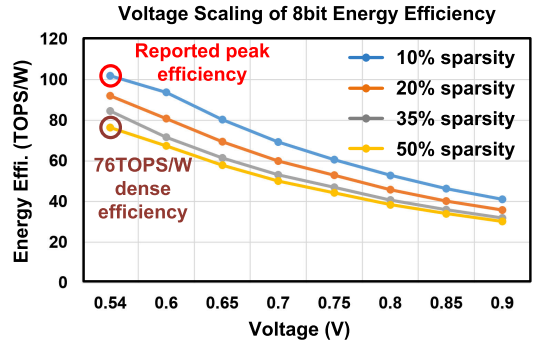
VI. Measurement Results

A. Chip Micrograph and Summary

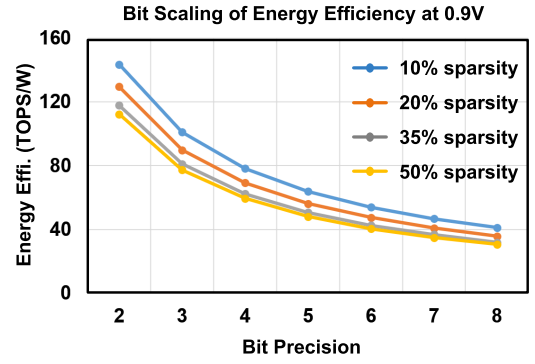
Fig. 14(a) presents the die photograph and summarizes the chip measurement results. Same as in [19], this chip is



(a)



(b)



(c)

Fig. 15. (a) Voltage–frequency scaling. (b) Voltage scaling of 8-bit energy efficiency. (c) Bit scaling of energy efficiency.

fabricated in 28-nm CMOS technology the CIM macro with an area of 0.028 mm^2. Fig. 14(b) demonstrates the test platform for the fabricated chip. The test chip is wire bonded to a SODIMM-profile adapter board and connected to the host PC for verification and measurements through an FPGA bridge. A customized compiler on the PC side generates instructions and data for chip configuration and execution. A current meter is inserted between the power supply and the test board for energy measurements.

As shown in Fig. 15(a), the voltage scaling of the chip is measured from 0.54 to 0.9 V. The operating frequency ranges from 20 to 240 MHz. The peak performance for 8-bit computation reaches 0.122 TOPS at 0.9 V, running at 240 MHz. Note that the calculation of throughput is different from the dot-product-based NN chips. In our multiply-less NN chip, a pair of comparisons and accumulation of one activation and one

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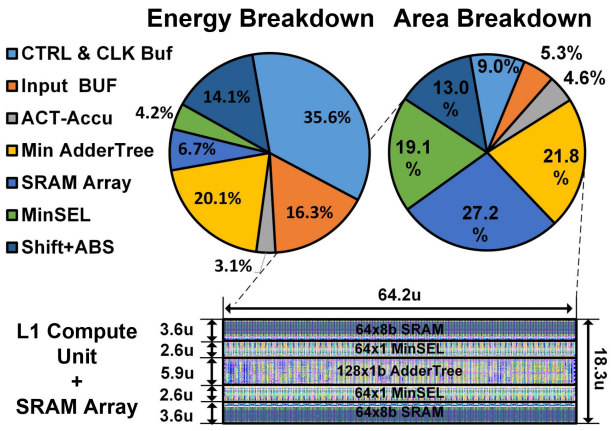


Fig. 16. Energy and area breakdown of test-chip.

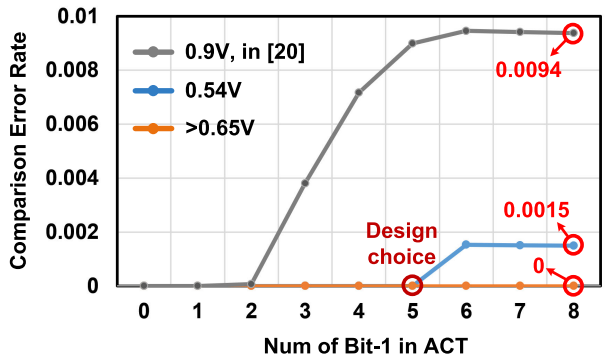


Fig. 17. Error rate from improved comparator and original comparator design in [19].

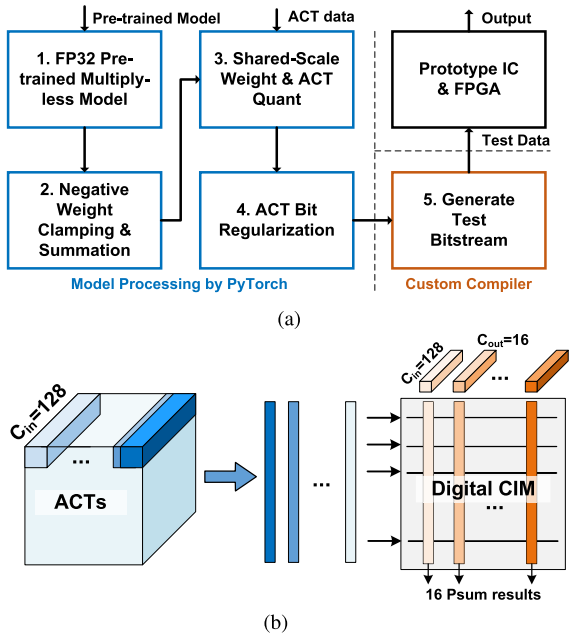


Fig. 18. (a) NN model deployment flow. (b) NN mapping method.

TABLE II
NN DEMONSTRATIONS

Dataset	CIFAR-10	ImageNet
NN Model	ResNet-20	ResNet-50
Bit Precision	8b/8b Weight/Activation	8b/8b Weight/Activation
Accuracy[1,2] (vs Float32)	91.74% (vs 91.84%)	74.8% (vs 74.9%)
Accuracy in [19]	91.55%	74.5%
Energy per Class (uJ)[1,2]	2.14	220.04
Throughput (fps)	3019.5	28.21

[1]. Off-chip memory access for activations and weights are excluded
[2]. Evaluated at 0.9V, 230MHz

weight is counted as two OPs. Since the bit-serial comparison scheme is adopted, it takes eight clock cycles to complete the two OPs. Fig. 15(b) presents the voltage scaling of 8-bit computation energy efficiency with different input sparsity. The highest energy efficiency is 102 TOPS/W at 0.54 V, 20 MHz, 10% input sparsity, and 50% weight sparsity, which is the same as in [19]. The bit scaling of energy efficiency at 0.9 V with different input sparsity is shown in Fig. 15(c). The linear bit scaling at higher bit-widths for energy efficiency demonstrates that the multiply-less digital CIM macro can overcome limitations imposed by the quadratic computational cost in the conventional MAC digital CIM macro.

Fig. 16 shows the energy breakdown of the entire macro at 0.9 V, including the control and clock buffer with 50% input and weight sparsity. The power breakdown shows that the L1 compute unit, which contains MinSEL, MinACCU, ActACCU, and ABS-ASM, consumes 40% overall energy. The area breakdown shows the L1 compute unit consumes 59% area. The area consumption of the SRAM banks, MinSEL, and ActACCU are almost identical, indicating that the area overhead of digital circuits is effectively reduced.

As mentioned, the charge-sharing problem, caused by the high ACT-W pairs, leads to comparison errors in the dynamic logic. We tested the comparator extensively and repeated them on different chips, the results of which are shown in Fig. 17. When the voltage is 0.54 V, the comparison error starts from six high ACT-W values. When the voltage exceeds 0.65 V, there is no comparison error. In contrast, the comparison error in [19] starts from two high ACT-W values and has a much higher error rate than this work. It firmly confirms that the proposed comparator in this article significantly improves the comparison accuracy. To alleviate this approximation error, we regularize the maximum number of "1"s in activations to 5 with lower bits truncated. The regularization is performed during NN inference with the help of potential element-wise digital peripherals. For example, the original activation is 63 (binary code: 00111111); after regularization, it becomes 62 (binary code: 00111110). Compared to regularizing the maximum number of "1"s to 3 in [19], this work reduces the number of activations that need to be regularized, thus reducing the regularization loss. For example, compared to [19] where 15% of the activations need to be regularized for ResNet-20 model, this work requires only 0.9% of the activations to be regularized, thus enabling higher accuracy inference.

B. Evaluation on Multiply-Less Neural Network

1) NN Model Deployment Flow: The macro architecture is co-designed with software mapping algorithms. To better

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TABLE III
COMPARISON WITH STATE-OF-THE-ART WORKS

	ISSCC'21 J. Su	ISSCC'22 P. Wu	ISSCC'21 Y. Chih	ISSCC'22 H. Fujiwara	ISSCC'22 B. Yan	ISSCC'22 D. Wang	This work
Technology	28nm	28nm	22nm	5nm	28nm	28nm	**28nm**
CIM Type	Analog CIM		Digital CIM				
NN Operand	Multiply + ADD					OR/AND+ ADD	**ABS+ADD**
Array Size	384Kb	1Mb	64kb	64kb	32kb	16kb	**16kb**
Cell Type	Compact 8T	Compact 6T	6T	12T	Compact 6T	8T	**Compact 8T**
Macro Area (mm^2)	N/A	N/A	0.202	0.0133	0.03	0.049	**0.028**
Voltage (V)	0.7-0.9	0.65-0.9	0.72	0.5-0.9	0.8	0.45-1.1	**0.54-0.90**
Input Bit	4/8b	4/8b	1b-8b	4b	1b-8b	1-4b	**2b-8b**
Weight Bit	4/8b	4/8b	4b	4b	1b/4b/8b	1b	**2b-8b**
8b Area Eff. (TOPS/mm^2)	N/A	N/A	4 (0.72V)	13.8 (0.5V)	0.18 (0.8V)	2.6 (0.9V)	**4.4 (0.9V)**
8b Energy Eff. (TOPS/W)	22.75	37.01	24.7 (0.72V)	63 (0.5V)	27.38 (0.8V)	15.5 (0.5V)	**102 (0.54V)**
Verified Dataset	Cifar-100		N/A			Cifar-10	**ImageNet[1,2]**

[1]: One comparsion and one accumulation are counted as two OPs
[2]: Measured at 0.54V, 20MHz, 10% input sparsity

outline the model deployment flow, Fig. 18(a) shows the custom off-line posttraining quantization toolchain for NN models used in this work. First, a floating-point multiply-less pretrained model is obtained using the standard training flow by PyTorch. Second, the negative weights are clamped to [0, $w_{\max}$] and merged to the WSUM, as mentioned in Section IV-A. Third, unlike the separate scaling factor for CNN, the multiply-less model requires the same scaling factors for activations and weights to perform the comparison. Meanwhile, more than 95% of weights and activations are in the range of 2^{-5}–2^{3} [35]. Therefore, a shared scaling factor for activations and weights quantization method is proposed to achieve the 8-bit quantization. Then, the activations are regularized with the maximum number of "1"s to 5, with lower bits truncated, to meet the needs of the improved comparator. Finally, the generated quantized model is deployed to the test chip for evaluation. Notably, this flow uses posttraining quantization without complex model retraining, which benefits various edge AI applications.

Given the relatively small size of the chip array, a layer-by-layer NN mapping method is adopted. During each layer's computation, as shown in Fig. 18(b), the 16 columns store the weights of 16 output channels, with each column storing the 128 weights of different input channels. During inference computation, 128 inputs are fed into the digital CIM macro and computed in parallel with the weights from the 16 columns. After eight clock cycles of comparison and accumulation, 16 Psum results are obtained.

2) NN Performance: Using this model deployment flow, several multiply-less models have been demonstrated on the prototype. Two models are presented as examples, including a ResNet-20 style multiply-less model for the CIFAR-10 dataset and a ResNet-50 style multiply-less model for the ImageNet dataset. In these models, the convolution multiplication is replaced by the L1-distance computation. As mentioned, we need to regularize the maximum number of "1"s in activations to 5 with lower bits truncated. Table II summarizes the results of these two benchmarks. The 8-bit ResNet-20 style multiply-less model for CIFAR-10 dataset achieves an energy efficiency of 2.14 μJ/image, and throughput of 3019.5 images/s, with a testing accuracy of 91.71%. The 8-bit ResNet-50 style multiply-less model for the ImageNet dataset achieves an energy efficiency of 220.04 μJ/image, and throughput of 28.21 images/s, with a testing accuracy of 74.8%. Compared to the 32-bit floating-point model, the loss of accuracy due to approximation and quantization is less than 0.1%, demonstrating the effectiveness of algorithm-to-circuit co-approximation. Compared to the 74.5% accuracy for the ImageNet dataset in [19], this work achieves higher accuracy by regularizing fewer activation values.

C. Comparison With State-of-the-Art Works

Table III compares this work with other state-of-the-art CIM macros. The peak area efficiency and energy efficiency are measured for 8-bit weights and activations. Compared with other CIM demonstrations, this work is the first CIM macro that supports the L1-distance computation to approximate the multiply-add operation. Meanwhile, this work achieves the highest area efficiency and energy efficiency compared to other works fabricated with similar technologies. Compared with state-of-the-art approximate digital CIM [15], this work achieves 6.58× energy efficiency and 1.7× area efficiency improvement while providing ImageNet validation. Compared to previous works, this work is the first design to achieve over 100-TOPS/W energy efficiency for 8-bit operation with ImageNet dataset validation.

VII. CONCLUSION

Digital CIM incorporates the processing unit into memory cells to reduce data movement energy and provide

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high bandwidth. However, bit-widths expansion introduces the quadratic cost due to multiplication, which limits further improvement for digital CIM. To address these challenges, an efficient compute scheme is proposed to approximate the multiply-add operations, thus guaranteeing linear area/energy scaling with bit-widths. This article demonstrates a prototype digital CIM macro in 28-nm CMOS based on a multiply-less NN and dynamic-logic-based approximate circuits. The macro achieves 102-TOPS/W peak energy efficiency and 4.4-TOPS/mm^2 peak area efficiency for 8-bit operations. Using the model deployment flow, several NN benchmarks are demonstrated. With the activation regularization and posttraining quantization techniques for approximation, the multiply-less models with 8-/8-b weights/activations show accuracy equivalent to floating-point counterparts. The fusion of digital CIM and approximation compute scheme provides new insights for algorithm-to-circuit co-design, with high energy efficiency for edge AI applications.

REFERENCES

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 2, pp. 84–90, Jun. 2012.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[3] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[4] A. Vaswani et al., "Attention is all you need," in *Proc. NIPS*, 2017, pp. 6000–6010.

[5] M. Horowitz, "1.1 Computing's energy problem (and what we can do about it)," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2014, pp. 10–14.

[6] J. Zhang, Z. Wang, and N. Verma, "In-memory computation of a machine-learning classifier in a standard 6T SRAM array," *IEEE J. Solid-State Circuits*, vol. 52, no. 4, pp. 915–924, Apr. 2017.

[7] M. Kang, S. K. Gonugondla, A. Patil, and N. R. Shanbhag, "A multi-functional in-memory inference processor using a standard 6T SRAM array," *IEEE J. Solid-State Circuits*, vol. 53, no. 2, pp. 642–655, Feb. 2018, doi: 10.1109/JSSC.2017.2782087.

[8] W.-S. Khwa et al., "A 65nm 4Kb algorithm-dependent computing-in-memory SRAM unit-macro with 2.3ns and 55.8TOPS/W fully parallel product-sum operation for binary DNN edge processors," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 496–498.

[9] X. Si et al., "24.5 A twin-8T SRAM computation-in-memory macro for multiple-bit CNN-based machine learning," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 396–398.

[10] H. Valavi, P. J. Ramadge, E. Nestler, and N. Verma, "A 64-tile 2.4-Mb in-memory-computing CNN accelerator employing charge-domain compute," *IEEE J. Solid-State Circuits*, vol. 54, no. 6, pp. 1789–1799, Jun. 2019.

[11] Z. Jiang, S. Yin, J.-S. Seo, and M. Seok, "C3SRAM: An in-memory-computing SRAM macro based on robust capacitive coupling computing mechanism," *IEEE J. Solid-State Circuits*, vol. 55, no. 7, pp. 1888–1897, Jul. 2020.

[12] H. Jia et al., "Scalable and programmable neural network inference accelerator based on in-memory computing," *IEEE J. Solid-State Circuits*, vol. 57, no. 1, pp. 198–211, Jan. 2022.

[13] H. Wang, R. Liu, R. Dorrance, D. Dasalukunte, D. Lake, and B. Carlton, "A charge domain SRAM compute-in-memory macro with C-2C ladder-based 8-bit MAC unit in 22-nm FinFET process for edge inference," *IEEE J. Solid-State Circuits*, vol. 58, no. 4, pp. 1037–1050, Apr. 2023, doi: 10.1109/JSSC.2022.3232601.

[14] H. Kim, T. Yoo, T. T. Kim, and B. Kim, "Colonnade: A reconfigurable SRAM-based digital bit-serial compute-in-memory macro for processing neural networks," *IEEE J. Solid-State Circuits*, vol. 56, no. 7, pp. 2221–2233, Jul. 2021.

[15] D. Wang, C. T. Lin, G. K. Chen, P. Knag, R. K. Krishnamurthy, and M. Seok, "DIMC: 2219TOPS/W 2569F^2/b digital in-memory computing macro in 28nm based on approximate arithmetic hardware," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, Sep. 2022, pp. 266–268.

[16] Y.-D. Chih et al., "16.4 An 89TOPS/W and 16.3TOPS/mm^2 all-digital SRAM-based full-precision compute-in memory macro in 22nm for machine-learning edge applications," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, vol. 64, Feb. 2021, pp. 252–254.

[17] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm^2 fully-digital computing-in-memory macro supporting wide-range dynamic-voltage-frequency scaling and simultaneous MAC and write operations," in *Proc. IEEE Int. Solid-State Circuits Conf.*, Feb. 2022, pp. 1–3.

[18] H. Mori et al., "A 4nm 6163-TOPS/W/b 4790-TOPS/mm^2/b SRAM based digital-computing-in-memory macro supporting bit-width flexibility and simultaneous MAC and weight update," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 132–134.

[19] Y. He et al., "7.3 A 28nm 38-to-102-TOPS/W 8b multiply-less approximate digital SRAM compute-in-memory macro for neural-network inference," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 130–132.

[20] B. Yan et al., "A 1.041-Mb/mm^2 27.38-TOPS/W signed-INT8 dynamic-logic-based ADC-less SRAM compute-in-memory macro in 28nm with reconfigurable bitwise operation for AI and embedded applications," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 188–190.

[21] J. Yue et al., "A 28nm 16.9–300TOPS/W computing-in-memory processor supporting floating-point NN inference/training with intensive-CIM sparse-digital architecture," in *Proc. IEEE Int. Solid-State Circuits Conf.*, Oct. 2023, pp. 1–3.

[22] S. Liu et al., "16.2 A 28nm 53.8TOPS/W 8b sparse transformer accelerator with in-memory butterfly zero skipper for unstructured-pruned NN and CIM-based local-attention-reusable engine," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 250–252.

[23] H. Diao et al., "A 28nm 128TFLOPS/W computing-in-memory engine supporting one-shot floating-point NN inference and on-device fine-tuning for edge AI," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2024, pp. 1–2.

[24] J. Yue et al., "STICKER-IM: A 65 nm computing-in-memory NN processor using block-wise sparsity optimization and inter/intra-macro data reuse," *IEEE J. Solid-State Circuits*, vol. 57, no. 8, pp. 2560–2573, Aug. 2022.

[25] J.-W. Su et al., "16.3 A 28nm 384kb 6T-SRAM computation-in-memory macro with 8b precision for AI edge chips," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 250–252.

[26] P.-C. Wu et al., "A 28nm 1Mb time-domain computing-in-memory 6T-SRAM macro with a 6.6ns latency, 1241GOPS and 37.01TOPS/W for 8b-MAC operations for edge-AI devices," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 1–3.

[27] J. Song et al., "A 4-bit calibration-free computing-in-memory macro with 3T1C current-programed dynamic-cascode Multi-Level-Cell eDRAM," *IEEE J. Solid-State Circuits*, vol. 59, no. 3, pp. 842–854, Mar. 2024.

[28] H. Chen et al., "AdderNet: Do we really need multiplications in deep learning?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1465–1474.

[29] H. Shu, J. Wang, H. Chen, L. Li, Y. Yang, and Y. Wang, "Adder attention for vision transformer," in *Proc. Adv. Neural Inf. Process. Systems(NIPS)*, vol. 34, Sep. 2021, pp. 19899–19909.

[30] X. Chen, C. Xu, M. Dong, C. Xu, and Y. Wang, "An empirical study of adder neural networks for object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 6894–6905.

[31] D. Song, Y. Wang, H. Chen, C. Xu, C. Xu, and D. Tao, "AdderSR: Towards energy efficient image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15643–15652.

[32] B. Li, Y. Xin, C. Li, Y. Bao, F. Meng, and Y. Liang, "AdderIC: Towards low computation cost image compression," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 2030–2034.

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[33] Y. Nie, K. Han, H. Diao, C. Liu, E. Wu, and Y. Wang, "Redistribution of weights and activations for AdderNet quantization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 22739–22751.
[34] H. Chen, Y. Wang, C. Xu, C. Xu, C. Xu, and T. Zhang, "Universal adder neural networks," 2021, *arXiv:2105.14202*.
[35] Y. Wang et al., "Addernet and its minimalist hardware design for energy-efficient artificial intelligence," 2021, *arXiv:2101.10015*.



Haikang Diao (Graduate Student Member, IEEE) received the B.Eng. degree in electronic engineering and the M.S. degree in circuits and systems from Fudan University, Shanghai, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree with the School of Integrated Circuits, Peking University, Beijing, China.

His research interests include computing-in-memory, model compression, and acceleration.



Yifan He (Graduate Student Member, IEEE) received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2019, where he is currently pursuing the Ph.D. degree.

His doctoral research includes emerging nonvolatile memories and in-memory computing designs with a focus on digital circuit technology for superior efficiency and compute density.

Mr. He was a recipient of the 2023–2024 IEEE Solid-State Circuits Society (SSCS) Predoctoral Achievement Award.



Xuan Li received the B.Eng. degree from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2015. He is currently pursuing the Ph.D. degree with Tsinghua University, Beijing, China.

He is also the Founder of Wiener Technology, Beijing. His research interests include energy-efficient data converter design, mixed-signal and analog signal processing system, and emerging sensing technology.



Chen Tang (Graduate Student Member, IEEE) received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include neural network compression and machine-learning accelerator.



Wenbin Jia received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2022, where he is currently pursuing the Ph.D. degree.

His research interests include computing-in-memory circuit design and hardware/software co-design for AI applications.



Jinshan Yue (Member, IEEE) received the B.S. and Ph.D. degrees from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2016 and 2021, respectively.

He is currently a Post-Doctoral Researcher and an Assistant Researcher with the Institute of Microelectronics of the Chinese Academy of Sciences, Beijing. His current research interests include energy-efficient neural network processor, nonvolatile memory, and computing-in-memory system.

Dr. Yue served as a TPC Member for IEEE A-SSCC2023 and ICCAD2023.



Haoyang Luo (Graduate Student Member, IEEE) received the B.S. degree from the School of Information and Electronics, Beijing Institute of Technology, Beijing, China, in 2021.

He is currently working toward the Ph.D. degree with the school of integrated circuits, Peking University, Beijing. His research interests include computing-in-memory, computing architecture, and hardware security.



Jiahao Song (Member, IEEE) received the B.Sc. degree (Hons.) from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2017, and the Ph.D. degree from Peking University, Beijing, China, in 2022.

He was a Post-Doctoral Researcher with Peking University, from 2022 to 2024. He is currently a Post-Doctoral Researcher with the University of California at San Diego, La Jolla, CA, USA. His research interests include memories, computing in-memory, and low-power mixed-signal circuits design.



Xueqing Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2007 and 2013, respectively.

He is currently an Associate Professor with the Department of Electronic Engineering, Tsinghua University. He has over 130 publications and holds 30 Chinese and U.S. patents. His research interests include mixed-signal circuit design, emerging memory, and memory-oriented computing exploration for AI acceleration.

Dr. Li is a Senior Member of CCF China. He received a few best paper awards, such as HPCA'15, ASP-DAC'17, and 2016 IEEE TMSCS, several teaching and thesis awards, and 2019 National Early-Career Award. He also serves on TPC in a few conferences, such as DAC, ICCAD, ASP-DAC, ISCAS, AICAS, GLSVLSI, ISVLSI, NVMSA, and COINS. He is an Associate Editor of the IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (TVLSI) SYSTEMS and IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING (TETC) and a Guest Editor of a few journal issues, such as *International Journal of Circuit Theory and Applications* and *Flexible and Printed Electronics*.

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Huazhong Yang received the B.S. degree in microelectronics and the M.S. and Ph.D. degrees in electronic engineering from Tsinghua University, Beijing, China, in 1989, 1993, and 1998, respectively.

In 1993, he joined the Department of Electronic Engineering, Tsinghua University, where he has been a Full Professor since 1998. He has been in charge of several projects, including projects sponsored by the National Science and Technology Major Project, the 863 Program, NSFC, and several international research projects. He has authored or co-authored over 500 technical articles and seven books, and holds over 180 granted Chinese patents. His research interests include wireless sensor networks, data converters, energy-harvesting circuits, nonvolatile processors, and brain-inspired computing.

Prof. Yang received the Distinguished Young Researcher by NSFC in 2000, the Cheung Kong Scholar by the Chinese Ministry of Education (CME) in 2012, the Science and Technology Award First Prize by the China Highway and Transportation Society in 2016, the Technological Invention Award First Prize by CME in 2019, the Gold Prize of iNEA 2019, and several best paper awards, including ISVLSI 2012, FPGA 2017, NVMSA 2017, and ASP-DAC 2019. He has served as the Chair for the Northern China ACM SIGDA Chapter Science in 2014, the General Co-Chair for ASP-DAC 2020, a Navigating Committee Member for AsianHOST'18, and a TPC Member for ASP-DAC 2005, APCCAS 2006, ICCCAS 2007, ASQED 2009, and ICGCS 2010.



Hongyang Jia (Member, IEEE) received the B.E. degree from Tsinghua University, Beijing, China, in 2014, and the M.A. and Ph.D. degrees from Princeton University, Princeton, NJ, USA, in 2016 and 2021, respectively.

From 2021 to 2022, he was a Research Post-Doctorate with NVIDIA, Santa Clara, CA, USA. Since August 2022, he has been with the Department of Electronic Engineering, Tsinghua University, where he is currently an Assistant Professor. His research focuses on advanced computing technologies and systems, rethinking how emerging computing fabrics can be integrated with architectures and algorithms to enhance system efficiency and performance. His primary research interests include approximate computing, mixed-signal computing, in-memory computing, energy-efficient circuits, and architectures for robotic computing and security applications.

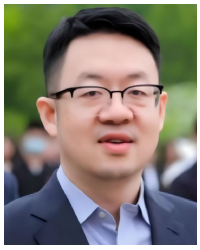
Prof. Jia received the 2017 Outstanding Student Designer Award from ADI Inc., and the 2022 Bede Liu Best Dissertation Award in Electrical and Computer Engineering from Princeton ECE. He serves on the technical program committees for the IEEE/ACM ICCAD Conference.



Yongpan Liu (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees from Tsinghua University, Beijing, China, in 1999, 2002, and 2007, respectively.

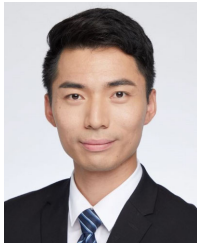
He is currently a tenured Full Professor with the Department of Electronic Engineering, Tsinghua University.

Dr. Liu is a Program Committee Member of ISSCC, ASSCC, and DAC. He received under 40 Young Innovators Award DAC 2017, Best Paper/Poster Award from ASPDAC 2021 and 2017, Micro Top Pick 2016, HPCA 2015, and Design Contest Awards of ISLPED in 2012, 2013, and 2019. He served as the General Secretary for ASPDAC 2021 and the Technical Program Chair for NVMSA 2019. He was an Associate Editor of the IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEM II: EXPRESS BRIEFS, and *IET Cyber-Physical Systems: Theory and Applications*. He also served as an A-SSCC2020 Tutorial Speaker and an IEEE CASS Distinguished Lecturer 2021.



Yuan Wang (Member, IEEE) received the Ph.D. degree in microelectronics from Peking University, Beijing, China, in 2006.

He is currently a Full Professor with the School of Integrated Circuits, Peking University. He has authored or co-authored more than 100 technical articles and conference contributions, including more than 30 articles published in IEEE TRANSACTIONS and journals. He has been granted over 70 patents. His current research interests include CMOS memory circuits, neuromorphic computing, in-memory computing, and spiking neural networks.



Xiyuan Tang (Senior Member, IEEE) received the B.Sc. degree (Hons.) from the School of Microelectronics, Shanghai Jiao Tong University, Shanghai, China, in 2012, and the M.S. and Ph.D. degrees in electrical engineering from The University of Texas at Austin, Austin, TX, USA, in 2014 and 2019, respectively.

He was a Design Engineer with Silicon Laboratories, Austin, from 2014 to 2017, where he was involved in the RF receiver design. From 2019 to 2021, he was a Post-Doctoral Researcher with The University of Texas at Austin. He is currently an Assistant Professor with Peking University, Beijing, China. His research interests include digitally assisted data converters, low-power mixed-signal circuits, and analog data processing.

Dr. Tang was a recipient of the IEEE Solid-State Circuits Society Rising Stars in 2020, Best Paper Award at Silicon Labs Tech Symposium in 2016, National Scholarship in 2011, and Shanghai Scholarship in 2010. He serves on the Technical Program Committees (TPC) of ISSCC. He also serves as an Associate Editor for IEEE SOLID-STATE CIRCUITS LETTERS.

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